

REPORT 2: Benchmarking and Deployment of a Hybrid LLM Architecture for Suicide Risk Detection in Australian Clinical Settings

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Abstract—Suicide remains a critical public health challenge in Australia, with 3,307 registered deaths in 2024 and an estimated 644,000 individuals experiencing serious suicidal thoughts annually. Existing clinical screening tools rely on static, episodic assessments that fail to capture real-time emotional dysregulation, motivating the need for scalable, always-available detection systems. This work investigates the use of Large Language Models (LLMs) for real-time suicide risk detection from conversational transcripts, targeting deployment within a privacy-compliant Australian clinical context.

A structured benchmark was conducted across six open-source models: MentalRoBERTa, MentaLLaMA-chat-7B, Mistral-7B, Llama-3.1-8B, Qwen2.5-7B, and IBM Granite 3.2-8B, evaluated on three publicly available datasets: Reddit C-SSRS, Human-AI Dialogue, and MHDialog. Models were compared using clinically grounded metrics including Quadratic Weighted Kappa (QWK), High-Risk Recall, Mean Absolute Error (MAE), Macro F1-score, and Inference Latency. IBM Granite 3.2-8B achieved the strongest ordinal alignment (QWK = 0.545, MAE = 1.19), while MentaLLaMA-chat-7B demonstrated the highest sensitivity for high-risk cases (Recall = 0.683).

A proof-of-concept evaluation on 10 curated clinical cases validated that structured prompting combining few-shot examples, Chain-of-Thought reasoning, and clinical safety constraints eliminates false negatives in high-acuity scenarios, achieving 100% High-Risk Recall. Based on these results, a hybrid architecture is proposed, selecting Llama-3.1-8B as the base model, enhanced through LoRA fine-tuning for domain adaptation, Chain-of-Thought prompting for structured clinical reasoning, and Retrieval-Augmented Generation (RAG) grounded in the Australian Suicide Assessment Kit (SAK). On-premise deployment on local GPU infrastructure ensures full compliance with the Australian Privacy Principles (APP) and positions the system as a responsible, clinician-supported early screening tool.

Index Terms—Large Language Models, Suicide Risk Detection, Clinical AI, Chain-of-Thought, RAG, Mental Health, C-SSRS, SAK, Australian Healthcare.

1 INTRODUCTION

SUICIDE remains a primary public health concern in Australia. In 2024, there were 3,307 registered deaths by suicide, representing an age-standardised rate of 11.8 per 100,000 residents [1]. Approximately 644,000 Australians have experienced suicidal thoughts or behaviour, and suicide is the leading cause of death for people aged 15 to 44 [2]. This highlights a clear need for better early detection tools.

Suicide remains a primary public health concern in Australia. In 2024, there were 3,307 registered deaths by suicide, representing an age-standardised rate of 11.8 per 100,000 residents [1]. Approximately 644,000 Australians have experienced suicidal thoughts or behaviour, and suicide is the leading cause of death for people aged 15 to 44 [2]. This highlights a clear need for better early detection tools, particularly those capable of operating at scale and in real-time. Beyond mortality statistics, the mental health system faces a structural demand gap, particularly in regional and under-resourced communities. Traditional assessment methods rely on episodic, face-to-face evaluations and often fail to capture short-term fluctuations in emotional state. Clinical evidence indicates that many individuals who die

by suicide were previously assessed as low risk [3], highlighting limitations in static checklist-based approaches and motivating the need for continuous, real-time monitoring systems.

Conversational AI offers a unique opportunity to address this gap. Unlike traditional face-to-face screening, an AI system can be available 24/7, reduces stigma, and can act as a first-line screening layer before referral to professionals. Large Language Models (LLMs), in particular, enable the analysis of naturalistic dialogue and the detection of subtle linguistic signals such as hopelessness, withdrawal, and implicit ideation. In February 2025, the Australian Government launched the National Suicide Prevention Strategy 2025–2035, which explicitly prioritises the integration of AI to improve accessibility and early intervention [2].

However, building a reliable suicide risk detection engine involves several technical challenges. These include linguistic ambiguity (e.g., distinguishing figurative expressions from genuine intent), the detection of implicit ideation where risk is not explicitly stated, and the need to model long-range conversational context across multiple dialogue turns. In addition, safety-critical deployment introduces a fundamental trade-off between false negatives and false positives, where missing a high-risk case has severe consequences. A recent scoping review found that while LLMs show promise for mental health applications, evaluation

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methods remain non-standardised and concerns about privacy and clinical validation are often overlooked [4].

The technical problem addressed in this work is the real-time classification of dialogue transcripts into tiered risk levels defined by Australian clinical standards. Specifically, this involves mapping unstructured conversational data into clinically meaningful categories such as low, moderate, and high suicide risk, aligned with established assessment frameworks.

The scope of this study focuses exclusively on the Detection Engine, the backend module that analyses transcripts and outputs both a tiered risk classification and an interpretable explanation of the underlying factors contributing to the decision. This includes identifying salient linguistic cues, contextual patterns, and behavioural indicators that support the model’s prediction, enabling transparency and alignment with clinical reasoning. It is assumed that a conversational agent already exists to generate the dialogue transcripts, and that any high-risk detection will trigger a human-in-the-loop referral protocol involving qualified mental health professionals. The system is therefore positioned as a decision-support and early screening tool rather than a replacement for clinical judgement, with outputs designed to be interpretable and to support transparent, clinically aligned decision-making.

The specific research objectives are fourfold. First, to establish a clinically grounded mapping between internationally recognised frameworks such as the C-SSRS [5] and Australian standards (SAK) [6] to LLM-based risk classification labels. Second, to evaluate the effectiveness of both specialised models (MentalRoBERTa) and open-source LLMs (MentalLaMA, LLaMA, Mistral, Qwen, and Granite) for suicide risk detection and interpretable reasoning across multiple conversational datasets, including Reddit Suicide-Watch (SWMH) [7], the Human-AI Dialogue Suicide Risk Dataset [8], and MHDIALOG [9]. Third, to investigate safety-critical performance trade-offs, prioritising recall (sensitivity) to minimise false negatives in high-risk categories, while also assessing complementary metrics such as latency, mean absolute error (MAE), and Appropriate Response Success Rate (ARSR). Fourth, to propose a privacy-compliant and clinically aligned system architecture for the secure deployment of LLM-based suicide risk detection systems within the Australian regulatory context.

This report is structured as follows. Section 2 presents the literature review. Section 3 benchmarks the candidate models. Section 4 outlines the system architecture and proof-of-concept. Section 5 discusses ethical considerations. Section 6 addresses limitations and future work, including alignment with the Sustainable Development Goals (SDGs), carbon footprint, and data security and privacy. Section 7 concludes the report.

2 LITERATURE REVIEW

This section reviews the evolution of natural language processing, from classical statistical methods to contemporary LLMs focused on clinical accuracy. Emphasis is placed on architectures that balance real-time triage efficiency with the interpretability required for high-stakes suicide risk detection.

2.1 Clinical Context: SAK and C-SSRS Frameworks

Any LLM-based detection system needs to align its outputs with established clinical frameworks. Two instruments are central to this project. The Suicide Assessment Kit (SAK) [6], developed at UNSW/NDARC, is a structured clinical triage tool used in mental health and alcohol and drug services. It classifies risk into levels such as no risk, ideation, intent, plan, and attempt. The Columbia-Suicide Severity Rating Scale (C-SSRS) [5] is a widely used and clinically validated tool for assessing suicidal thoughts and behaviours, and is commonly applied in both research and clinical practice.

Together, these frameworks provide a common way to represent suicide risk for the proposed detection engine. In this work, SAK-based categories are mapped to an ordinal structure inspired by C-SSRS severity levels [?], allowing consistent evaluation of risk across both ideation intensity and behavioural indicators.

Recent work [11] has explored the use of C-SSRS-style annotations to evaluate how well LLMs can detect and reason about suicide risk in social media data such as Reddit SuicideWatch posts.

2.2 Taxonomy of Mental Health LLMs

Mental health language models can be broadly grouped into encoder-based models and decoder-based generative models. These differ in architecture, capability, and suitability for clinical applications.

2.2.1 Encoder Models

The introduction of BERT (Bidirectional Encoder Representations from Transformers) marked a significant advance for text classification in mental health applications. Unlike previous recurrent neural networks, BERT-based models process text bidirectionally, capturing context from both left and right [12]. RoBERTa, an improved variant with more robust training, established strong baselines for risk classification from clinical text and social media data [13].

Domain-specific encoder variants such as MentalBERT and MentalRoBERTa [14] are pre-trained on approximately 15 million sentences from mental health-related Reddit communities. These models achieve strong performance on explicit suicide detection tasks, reaching peak F1 scores of approximately 97.4% across datasets such as SWMH, Dread-it, SAD, and T-SID. MentalRoBERTa-Caps [15] extends this architecture with capsule network layers, improving efficiency by reducing inference time while maintaining competitive accuracy, making it suitable for deployment scenarios with latency constraints.

2.2.2 Decoder-based generative Models

Decoder-based generative models are autoregressive systems that generate text token-by-token. Unlike encoder models, they are capable of producing explanations and structured reasoning, which is particularly relevant for clinical decision-support systems.

Commercial LLMs.

Models such as GPT-3.5, GPT-4, GPT-4o (OpenAI), Claude (Anthropic), and Gemini (Google) represent the most advanced proprietary systems. These models show strong performance in zero-shot clinical reasoning tasks

and tend to align more closely with human annotations in suicide risk classification tasks [16]. [10]. However, their use is constrained by cloud-based deployment, which raises privacy and data sovereignty concerns in regulated healthcare contexts such as Australia [17].

Open-source LLMs.

The ecosystem of open-source Large Language Models (LLMs) has expanded significantly, offering specialized architectures such as Meta’s LLaMA series (including LLaMA 2 and 3), Mistral AI’s dense and MoE variants (Mistral 7B and Mixtral 8x22B), and Google’s Gemma family. Furthermore, high-reasoning models like Alibaba’s Qwen3 and DeepSeek-R1, alongside enterprise-grade solutions like IBM’s Granite 4.1, provide an alternative paradigm focused on local deployment and model transparency. In mental health applications, studies show that open-source models can achieve competitive performance when properly prompted or fine-tuned, although they generally underperform commercial models in complex reasoning tasks [10].

Domain-adapted variants such as MentalLLaMA [18], MedGemma, and related mental-health-tuned models further improve performance by incorporating instruction tuning and domain-specific datasets.

2.3 Key Architectural Concepts

Model performance in mental health applications is influenced by several architectural factors.

Model scale. Model scale varies significantly across modern LLMs, ranging from small encoder models (e.g., BERT-base with 110M parameters) to large-scale decoder models exceeding 70B parameters such as LLaMA 3.3-70B and proprietary systems like GPT-4-class models. In general, larger models demonstrate improved reasoning ability, particularly in multi-step and ambiguous decision-making tasks such as implicit suicide ideation detection [11]. However, performance gains are not strictly linear, and smaller models can still perform competitively in constrained classification settings.

Prompt engineering. plays a critical role in decoder-based LLM performance. Techniques such as few-shot learning, chain-of-thought (CoT), self-consistency (SC-CoT), and tree-of-thought (ToT) improve reasoning performance by guiding intermediate inference steps rather than relying on direct classification [19]. In mental health applications, CoT prompting has been shown to improve both classification accuracy and interpretability by making the reasoning process partially observable.

Alignment methods. Modern decoder-based LLMs rely on a combination of supervised fine-tuning (SFT), instruction tuning, and reinforcement learning from human feed-

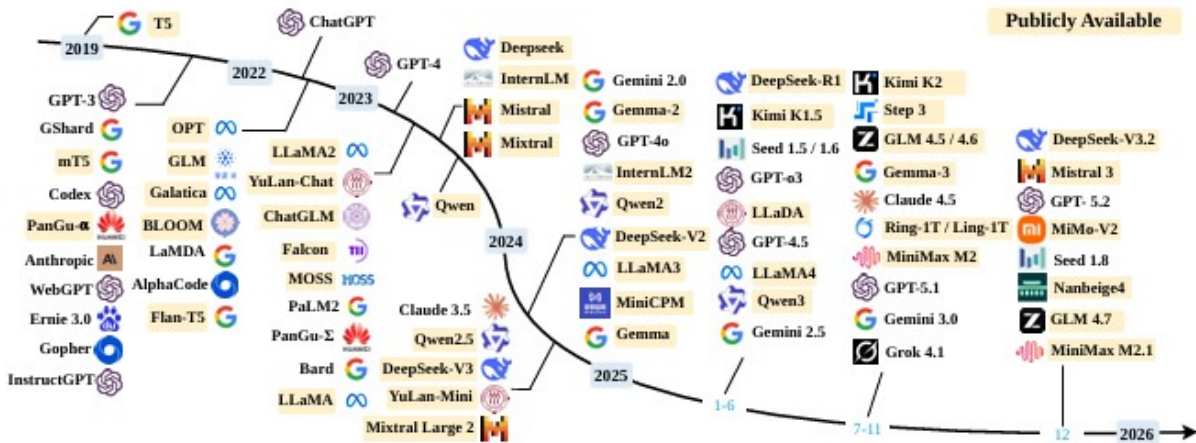


Fig. 1. Timeline of representative LLMs released in recent years. Models with publicly available checkpoints are highlighted in yellow. Source: [10]

TABLE 1
Comparison of LLM families for mental health risk detection systems

Model Type	Strengths	Limitations	Deployment Fit
Encoder Models	Fast inference, strong performance on explicit suicide risk detection	Limited interpretability and poor handling of implicit ideation	First-stage filtering in real-time systems
Commercial LLMs	Strong reasoning, high alignment with clinical judgement	Cloud-only, privacy risks, high cost, regulatory (APP) constraints	Not for sensitive or on-premise deployment
Open-source LLMs	On-premise deployment, data sovereignty, privacy-preserving	Variable reasoning quality, sensitive to prompting strategies	Regulated environments requiring local inference

back (RLHF) to improve alignment with human intent and safety requirements [20]. These techniques significantly enhance performance in conversational and safety-critical tasks such as mental health screening.

Explainability. In LLMs is closely related to reasoning-based prompting techniques such as chain-of-thought generation. Unlike encoder models, decoder-based models can produce intermediate reasoning steps that approximate clinical justification. However, these explanations are not guaranteed to be faithful to the internal decision process, which limits their reliability in high-stakes clinical settings [20].

2.4 Comparison of Methods and Trade-offs

Across the literature, model performance is typically evaluated using standard classification metrics such as precision, recall, and F1-score for suicide risk detection tasks [16]. In safety-critical settings, recall is often prioritised to reduce the risk of false negatives. Some studies also report ordinal-aware metrics such as Mean Absolute Error (MAE), particularly when evaluating severity-based frameworks like C-SSRS, as well as latency for real-time or conversational applications.

Different model families present distinct trade-offs between performance, interpretability, privacy, and deployability, which are summarised in Table 1.

2.5 Challenges and Research GAP

Despite progress in LLM-based mental health systems, several challenges remain.

First, most models still struggle with implicit suicidal ideation, particularly when language is indirect or metaphorical. This is especially relevant in conversational settings, where risk signals may appear gradually rather than in explicit statements.

Second, evaluation frameworks are inconsistent across studies, making direct comparison of results difficult. This limits the ability to establish reliable benchmarks for real-world deployment of suicide risk detection systems.

Third, there is a lack of deployment-focused research in Australian healthcare settings, particularly under privacy constraints defined by national guidelines. This creates a gap between research prototypes and systems that can be safely integrated into local clinical workflows, especially in scenarios requiring on-premise processing and human-in-the-loop escalation.

Finally, most systems optimise for classification performance rather than safety-critical recall, which is essential in suicide prevention contexts where false negatives are highly critical. In practical terms, this means that high-risk cases must be prioritised for detection and referral rather than balanced accuracy metrics.

These limitations directly motivate the development of a localised LLM-based detection engine designed for conversational transcripts, with a focus on early screening, risk stratification, and referral support rather than autonomous clinical decision-making.

3 BENCHMARKING RESULTS

To identify the most suitable LLM-based architecture for the proposed suicide risk detection engine, the candidate models are compared in terms of classification accuracy, recall

(sensitivity), and explainability, which are critical requirements for clinical safety and real-time triage applications.

3.1 Datasets

The benchmarking experiments use publicly available datasets commonly employed in suicide risk detection research. Three datasets were selected to enable model development under different data conditions.

3.1.1 Reddit SuicideWatch and Mental Health Collection (SWMH)

The primary dataset used in this study is the Reddit SuicideWatch and Mental Health Collection (SWMH) [7], one of the largest publicly available datasets for suicide ideation detection. The dataset contains approximately 54,412 posts collected from mental health-related subreddits such as r/SuicideWatch, r/depression, and r/anxiety. Each post is labelled with multi-class annotations for suicidal ideation and mental disorders. The dataset is pre-divided into train, validation, and test splits, making it suitable for supervised fine-tuning of transformer-based models. Its primary limitation is that Reddit posts are monologues rather than conversational dialogues, and labels are self-reported without clinical validation.

3.1.2 MHDialog Dataset

To complement SWMH with conversational data, the MHDialog dataset [9] is included. The MHDialog dataset [21] is designed for benchmarking mental health conversational AI systems. It contains multi-turn dialogues between users expressing psychological distress and AI assistants providing responses.

Unlike Reddit-based datasets, MHDialog focuses on conversational structure, where risk signals may appear gradually across multiple turns. This makes it more suitable for evaluating dialogue-based systems.

The dataset is part of the MHDash platform, which evaluates not only classification performance but also response quality, including aspects such as empathy, safety, and helpfulness.

3.1.3 Human-AI-Dialogue-Suicide-Risk-Dataset

The Human-AI-Dialogue-Suicide-Risk-Dataset [8] is a recent dataset that contains 4,040 annotated human-AI dialogue samples focused on mental health and suicide risk assessment. Each sample consists of a multi-turn conversation where the user expresses distress and the AI provides responses. The dataset includes risk-related annotations that reflect different levels of suicide severity: ideation (thoughts of dying without specific plan), behavior (actions suggesting increased risk), indicator (signs of severe distress), attempt (past attempts or imminent plans), and safe (no immediate risk).

A small proof-of-concept evaluation is conducted using a subset of 10 additional cases inspired by this dataset. These cases are designed to simulate realistic scenarios and evaluate model behaviour in controlled conditions.

3.2 Performance metrics

In the domain of mental health, lexical similarity metrics such as BLEU or ROUGE are insufficient, as they fail to capture clinical risk semantics [?]. Consequently, an evaluation framework based on patient safety and technical viability is defined.

3.2.1 Quadratic Weighted Kappa (QWK)

This is the primary metric for evaluating the alignment with the C-SSRS scale (0–6). Since the scale is ordinal, QWK measures the agreement between the model and the human expert, penalizing distance errors more severely (e.g., misclassifying a level 0 as a 5) than minor adjacent errors.

$$\kappa_w = 1 - \frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}} \quad (1)$$

Where $w_{i,j}$ represents the squared weight of the error between categories i and j [cite: 1].

3.2.2 Recall High-Risk (Sensitivity)

In mental health applications, a “false negative” can have fatal consequences. **Recall HR** measures the model’s effectiveness in identifying critical cases with a score ≥ 4 (suicidal ideation with intent or plan)[cite: 1]. We prioritize high Recall to ensure no life-threatening situation is overlooked.

$$\text{Recall}_{HR} = \frac{TP_{HR}}{TP_{HR} + FN_{HR}} \quad (2)$$

3.2.3 Mean Absolute Error (MAE)

The **MAE** indicates the average number of levels the model deviates from the ground truth on the C-SSRS scale[cite: 1]. A low MAE (as seen in **Granite 3.2**) ensures that predictions remain clinically grounded and close to the actual risk level.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

3.2.4 Macro F1-Score

Given that mental health datasets are often imbalanced (with a higher frequency of low-risk samples), we employ the **Macro F1-Score** to evaluate performance fairly across all risk levels, preventing the majority class from inflating the overall results.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

3.2.5 Inference Latency

Latency measures the total response time from the user’s input. In crisis situations, responses must be immediate; a maximum threshold of 3 seconds is defined to maintain conversational flow and ensure timely intervention.

$$\text{Latency} = T_{\text{final}} - T_{\text{start}} \quad (5)$$

3.2.6 Appropriate Response Success Rate (ARSR)

This metric evaluates the **explainability** and the quality of the generated output. It measures the model’s capacity to provide a coherent clinical justification for its detection, accompanied by an empathetic tone and validated crisis resources.

$$\text{ARSR} = \frac{\text{Clinically Appropriate Responses}}{\text{Total Evaluations}} \quad (6)$$

Although the primary benchmark focuses on quantitative classification, the proposed methodology incorporates the **ARSR** to evaluate the clinical quality of the generated output.

3.3 Model Selection and Configuration

A structured benchmark was conducted on six open-source models to identify the most robust backbone for suicide risk detection: MentalRoBERTa, MentaLLaMA-chat-7B, Mistral-7B-Instruct-v0.2, Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, and IBM Granite 3.2-8B-Instruct. These candidates were selected to ensure compliance with Australian data privacy standards, as open-source weights allow for complete data sovereignty by avoiding third-party API dependencies. The models were chosen for their balanced parameter size (125M to 8B), allowing deployment on local infrastructure, specifically an NVIDIA RTX 3060 (12GB VRAM) using 4-bit quantization, while maintaining high clinical performance.

The evaluation followed a Zero-Shot paradigm, meaning no fine-tuning was performed on the test datasets to assess the models ability to generalize the C-SSRS framework. The selection covers four technical approaches: MentalRoBERTa as a domain-specific encoder baseline using NLI-based inference; MentaLLaMA to test specialized psychiatric fine-tuning; Llama-3.1, Mistral, and Qwen2.5 as foundational LLMs for zero-shot reasoning; and IBM Granite 3.2 for its native Chain-of-Thought (CoT) capabilities, which are essential for clinical explainability.

3.4 Benchmarking Results

The evaluation was performed across three diverse annotated datasets: HumanAI Dialogs (N=100), Reddit C-SSRS (N=100), and MHDialog (N=92). Due to limited computational resources, the sample size was restricted to approximately 100 entries per dataset, with a pre-processing pipeline truncating texts to 3,000 characters to fit context windows.

3.4.1 Quantitative Analysis

Models were evaluated using the primary metrics previously defined: Quadratic Weighted Kappa (QWK), Recall (High-Risk), Mean Absolute Error (MAE), Macro F1-Score, and Inference Latency. This analysis focuses on the statistical alignment between model predictions and ground-truth clinical labels, identifying which architecture minimizes critical underestimations of risk.

The benchmark results reveal a significant trade-off between specialized sensitivity and overall architectural alignment. IBM Granite 3.2-8B emerged as the most robust candidate, achieving the highest QWK (0.545) and the lowest

TABLE 2
Quantitative Performance Benchmark of LLM Candidates

Model	QWK	Rec. HR	MAE	F1	Lat.(s)
Granite 3.2-8B	0.545	0.381	1.19	0.276	10.76
Llama-3.1-8B	0.499	0.237	1.31	0.233	4.12
Qwen 2.5-7B	0.477	0.131	1.25	0.237	2.97
MentaLLaMA	0.417	0.683	1.50	0.138	6.25
Mistral-7B	0.419	0.062	1.33	0.237	3.61
M-RoBERTa	0.134	0.587	1.97	0.154	0.01

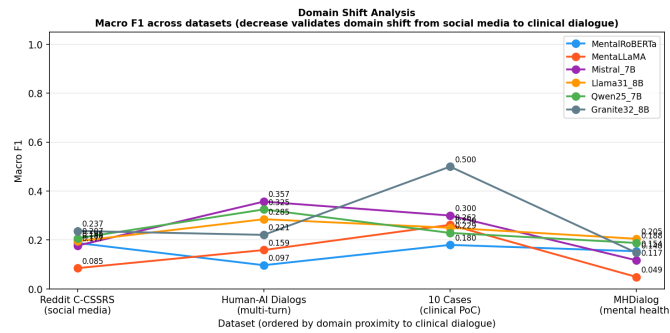


Fig. 2. Macro F1 across Datasets.

MAE (1.19). This suggests that its native Chain-of-Thought capabilities allow for a more precise calibration across the entire C-SSRS ordinal scale, avoiding the extreme misclassifications that penalize weighted kappa scores.

In contrast, MentaLLaMA-chat-7B exhibited the highest Recall for High-Risk cases (0.683), significantly outperforming the foundational models in safety sensitivity. However, its lower QWK and higher MAE indicate a tendency toward over-prediction, often labeling moderate cases as high-risk. While this safety-first bias is preferable to under-detection in clinical settings, it may lead to alert fatigue in real-time monitoring.

The foundational models (Llama-3.1, Qwen 2.5, and Mistral) showed moderate performance, with Llama-3.1-8B leading the group in QWK (0.499). However, they generally struggled with High-Risk Recall in a zero-shot setting, likely due to safety filters that can trigger refusals or neutral responses when encountering explicit suicidal ideation.

Finally, while MentalRoBERTa offers instantaneous inference (0.01s), its poor alignment with the C-SSRS scale (QWK 0.134) renders it insufficient as a primary reasoning engine.

3.4.2 Qualitative Analysis

A full-scale qualitative evaluation using an LLM-as-a-Judge framework (e.g., ARSR) was not conducted across the entire benchmark due to the high computational and API cost associated with evaluating large-scale conversational datasets. This step is therefore considered as future work.

Instead, a targeted manual evaluation was performed on a subset of responses from the Human-AI-Dialogue-Suicide-Risk-Dataset. This approach allows a preliminary assessment of explainability and clinical response quality while remaining feasible within resource constraints.

Each model output was evaluated based on three criteria: clinical justification, empathetic tone, and inclusion of safety-oriented reasoning. These criteria were used to approximate the Appropriate Response Success Rate (ARSR).

TABLE 3
Qualitative evaluation using ARSR on Human-AI examples

Model	Clinical	Empathy	Safe	ARSR
IBM Granite 3.2	1	0	1	0.67
LLaMA 3.1-8B	1	0	1	0.67
Qwen 2.5-7B	1	0	0	0.33
Mistral-7B	1	0	0	0.33
MentaLLaMA	0	0	0	0.00
MentalRoBERTa	0	0	0	0.00

The qualitative results highlight clear differences in explainability across model types. IBM Granite and LLaMA 3.1 demonstrate the strongest performance, providing structured clinical reasoning and identifying key risk indicators such as intent, persistence of ideation, and contextual stressors.

In contrast, models such as Mistral and Qwen identify relevant signals but produce shorter and less comprehensive explanations, reducing their effectiveness for clinical interpretation.

MentaLLaMA shows limited coherence in its justification, while MentalRoBERTa, as an encoder-based model, does not generate explanations at all, resulting in an ARSR of zero.

Overall, these results suggest that models with stronger reasoning capabilities not only achieve better quantitative alignment (QWK and MAE), but also produce more clinically useful explanations, reinforcing the importance of explainability in safety-critical mental health applications.

4 METHODOLOGY

This section details the experimental design and technical architecture of the proposed detection system. It describes the development of a hybrid framework—integrating LoRA fine-tuning, Chain-of-Thought reasoning, and Retrieval-Augmented Generation (RAG)—to ensure clinical safety and regulatory compliance within the Australian healthcare context.

4.1 System Overview

The proposed system is designed as a clinical decision-support pipeline integrated within a broader conversational AI ecosystem.

The system assumes the existence of an upstream conversational agent that generates user dialogue. These transcripts are passed to the Detection Engine, which processes them through a layered architecture.

At the core of the system, a Monitoring Loop continuously analyses each dialogue turn in real time. This loop triggers a Clinical Evidence Extraction (CEE) module, which identifies salient linguistic cues such as expressions of intent, plans, emotional distress, and behavioural indicators.

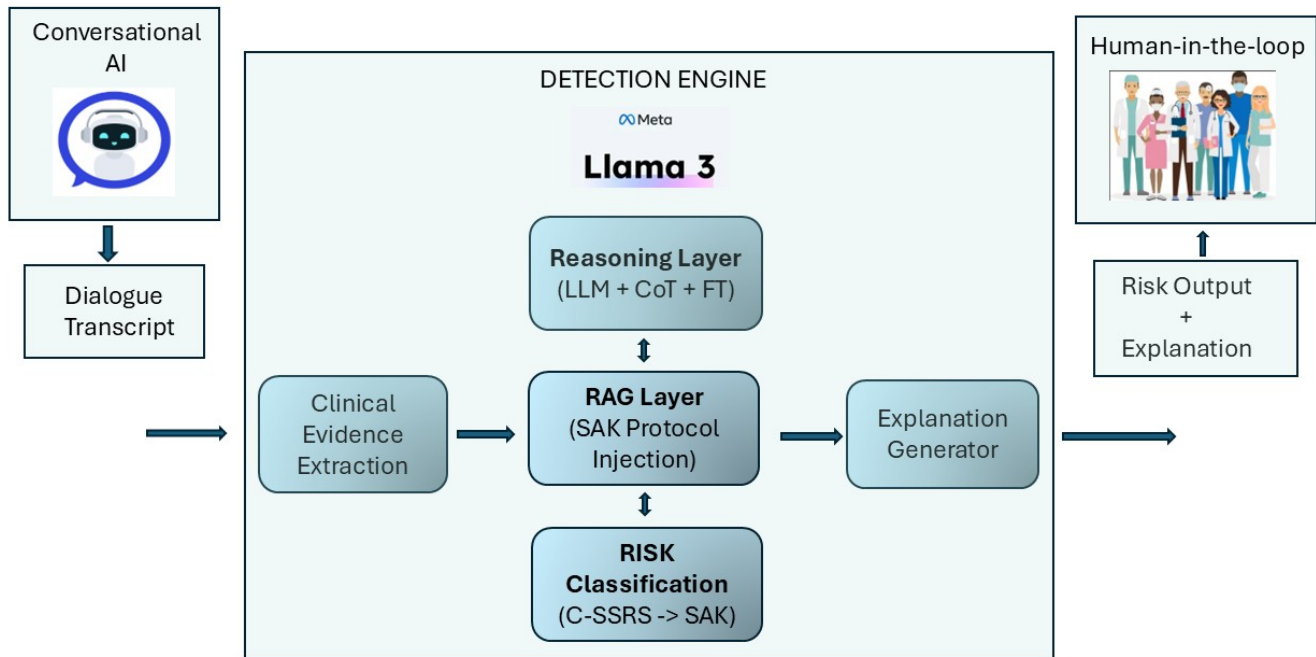


Fig. 3. System Overview - Suicide Risk Detection.

These extracted signals are then processed by a Reasoning Engine, based on a Large Language Model (LLM), which performs structured reasoning to map the dialogue into clinically meaningful categories aligned with the Columbia-Suicide Severity Rating Scale (C-SSRS). The outputs are further mapped to the Australian Suicide Assessment Kit (SAK) for local clinical alignment. A Retrieval-Augmented Generation (RAG) layer injects this protocol up-to-date into the reasoning process, ensuring that predictions are grounded in official guidelines.

Finally, the system outputs: A tiered suicide risk classification with a transparent explanation of the reasoning behind the decision.

In high-risk scenarios, the system triggers a human-in-the-loop escalation, enabling timely intervention by qualified mental health professionals. The system is therefore positioned as an early screening and decision-support tool rather than a replacement for clinical judgement.

Figure X illustrates the overall system architecture, including the monitoring loop, evidence extraction module, and escalation workflow.

4.2 Model Selection Strategy

The model selection process is guided by the benchmarking results and proof-of-concept evaluation presented in the previous section. Multiple factors were considered, including quantitative performance (QWK, Recall, MAE), latency, model size, and qualitative reasoning capability.

The previous benchmark revealed that: IBM Granite demonstrated strong reasoning and alignment with clinical scales and MentaLLaMA showed high sensitivity for detecting high-risk cases.

However, neither model provided the optimal balance between reasoning, controllability, deployment feasibility, and adaptability.

Therefore, the final decision was to adopt Llama 3.1-8B Instruct as the base model. This choice reflects a strategic trade-off: selecting a general-purpose model with strong instruction-following capabilities that can be further specialised.

The choice of Llama 3.1-8B specifically targets the model's superior instruction-following capabilities. The goal is to integrate the reasoning strengths observed in Granite and the sensitivity of MentaLLaMA into a controllable and deployable architecture.

4.3 Proof of Concept: Prompt Engineering Validation

Before proceeding to full-scale training, a proof of concept (PoC) was conducted using the 10 curated clinical cases to evaluate the selected Llama 3.1-8B model behaviour under different prompting strategies, mapping to SAK-Aligned Suicide Risk Assessment.

S1: Zero-shot baseline. The model attempted classification without guidance, resulting in critical failures, including false negatives in high-risk cases.

S2: Few-shot baseline. Introducing labelled examples improved consistency but lacked structured reasoning.

S3: Few-shot + Chain-of-Thought (CoT). The model was guided to explicitly reason step-by-step (e.g., identifying intent, plan, behavioural indicators), improving interpretability and accuracy.

S4: Few-shot + CoT + Safety Constraints. A clinical rule-based override was introduced, if SAK Item 8 (specific plan)

is detected, the model must classify the case as high risk. This eliminated false negatives and achieved 100% recall in the PoC.

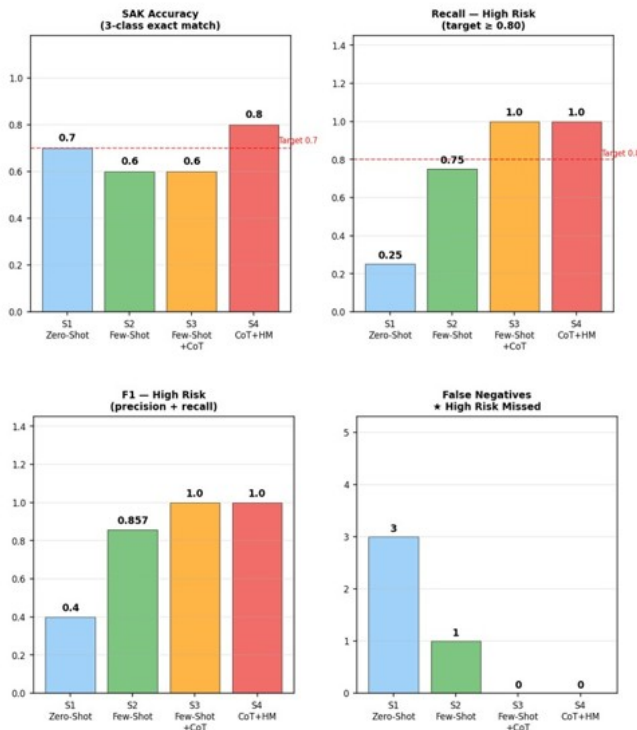


Fig. 4. Proof of Concept: Prompt Engineering Validation.

This PoC validates that structured reasoning combined with safety constraints can significantly reduce false negatives in high-risk scenarios.

The PoC and benchmark results allow us decide that the base model (Llama 3.1-8B Instruct) will be customised through three key mechanisms: LoRA fine-tuning for domain adaptation, Chain-of-Thought (CoT) prompting for structured reasoning, and Retrieval-Augmented Generation (RAG) to ground predictions in clinical and specific protocols.

4.4 Hybrid Architecture: Fine-Tuning + RAG

To move beyond static prompting, the system adopts a hybrid architecture combining fine-tuning, CoT and retrieval-based reasoning.

Fine-tuning (Reasoning Layer). LoRA fine-tuning is applied to Llama 3.1-8B to adapt the model to the mental health domain with different datasets. The objective is not only to improve classification accuracy but to internalise structured clinical reasoning patterns. The model is trained using Chain-of-Thought examples to replicate the reasoning behaviour observed in stronger models such as Granite, while incorporating domain-specific sensitivity similar to MentaLLaMA.

Retrieval-Augmented Generation (RAG). The RAG layer addresses the limitations observed during the PoC, where safety rules embedded in prompts improved performance and explainability based on evidence. Instead of relying on static prompts, the system retrieves relevant

sections of the Suicide Assessment Kit (SAK) and injects them into the model context. This ensures that predictions are grounded in explicit clinical evidence and reduces hallucinations.

This separation between reasoning (learned) and knowledge (retrieved) provides a robust mechanism to reduce hallucinations, improve explainability and maintain regulatory alignment.

4.5 Dataset Considerations

The model is trained using a combination of datasets introduced in the benchmarking phase, including SWMH, MHDIALOG, and Human-AI Dialogue datasets. Additional datasets such as CRADLE and MentalChat16K are incorporated to enrich conversational diversity and improve reasoning capabilities.

Class imbalance and variability across datasets are mitigated through preprocessing and balanced sampling strategies, ensuring robust learning across different risk levels and linguistic styles.

4.6 Implementation and Deployment Considerations

The system is designed to be both high-performing and deployable within clinical environments.

By utilizing 4-bit NF4 quantization, the inference engine can be hosted on localized, medical-grade GPU infrastructure (e.g., NVIDIA A10/RTX 6000 series). This ensures full compliance with APP 8 regarding data sovereignty, as no patient transcripts ever leave the hospital's private network.

Model optimisation. The system uses 4-bit quantisation (NF4) to reduce memory requirements, enabling deployment on mid-range GPUs while maintaining performance.

On-premise deployment. All processing is performed locally to comply with Australian Privacy Principles (APP), ensuring that sensitive patient data does not leave the system.

Scalability. Different size of model variants (e.g., 8B,3B parameters) will be explored to enable deployment on edge devices while preserving acceptable reasoning performance.

Regulatory compliance. The system is designed as Software as a Medical Device, aligning with Australian regulatory frameworks for clinical AI systems.

Security. No patient transcripts ever leave the hospital's private network without strict user authorization. Data protection mechanisms such as encryption will be incorporated to ensure compliance with healthcare data standards.

This methodology prioritises safety-critical recall, explainability, and regulatory compliance, aligning technical design with real-world clinical requirements.

5 AI ETHICS

The Australian Government has defined eight Artificial Intelligence Ethics Principles to guide the responsible design and deployment of AI systems [22]. These principles are particularly important for LLMs development.

This project considers these principles when designing the suicide risk detection engine, as they are essential to ensure responsible deployment.

5.1 Human, Social and Environmental Wellbeing

The system's primary objective is harm reduction. Every design choice (100% Recall mandate, on-premise data storage, clinician-in-the-loop) is subordinated to the wellbeing of vulnerable users. Additionally, using QLoRA efficiency for fine-tuning minimizes the carbon footprint of the project compared to full-parameter training, aligning with environmental sustainability goals.

5.2 Human-Centred Values

The engine is designed as a screening assistant, not a replacement for clinical judgment. In line with our architecture, the AI never communicates a risk level directly to the user; instead, it surfaces evidence-based alerts to a clinician dashboard. This ensures that a human-in-the-loop is always responsible for the final intervention, preserving the human-centred nature of mental care. User autonomy is preserved through informed consent at session start.

5.3 Fairness

We acknowledge that base models like Llama 3.1 can carry demographic biases toward English-speaking, Western, young-adult populations. The Australian clinical corpus fine-tuning phase specifically targets under-represented groups: Aboriginal and Torres Strait Islander communities, LGBTQ+ youth, and regional/remote users. To counter this, our methodology includes a specialized fine-tuning phase using the differents dataset, which includes clinician-annotated examples that help the model distinguish between diverse expressions of distress. Our evaluation focuses on Macro F1-scores to ensure the model performs equitably across different risk categories, not just the majority, trying mitigate this bias.

5.4 Privacy Protection and Security

Adhering to APP 8 and APP 11, the system is built for on-premise deployment within Australian AWS zones. The technical choice of an 8B parameter model allows for local inference, ensuring that sensitive patient transcripts never leave the secure clinical environment. Transcripts are processed in memory without persistent storage unless explicit consent is obtained (APP 3).

5.5 Reliability and Safety

Safety is technically enforced through our Hybrid RAG-FT architecture. By separating Reasoning (Fine-Tuning) from Clinical Knowledge (RAG), we mitigate the risk of hallucinations. The model is forced to cite the SAK protocol verbatim from the retrieved context, ensuring that its classification is always grounded in verified Australian clinical standards rather than imagined risk factors.

5.6 Transparency and Explainability

Consistent with our Chain-of-Thought (CoT) methodology, the system generates a clinical rationale for every classification. This rationale is logged in the clinician dashboard, enabling the reviewing clinician to understand exactly why the alert was generated. This transparency allows clinicians to quickly audit the AI's reasoning and understand the evidence behind every high-risk notification.

5.7 Contestability

Clinicians can override any risk classification via the dashboard with a documented reason. Users can request review of any automated crisis response. All overrides are logged and used to improve subsequent fine-tuning batches, creating a continuous improvement loop grounded in expert clinical judgment.

5.8 Accountability

Responsibility is shared and documented: the AI startup is accountable for model accuracy and security; the partner health service is accountable for clinical protocols and escalation; by registering the system as Software as a Medical Device (SaMD) under the TGA, we establish a clear framework of regulatory accountability for the system's clinical performance and safety.

6 DISCUSSION

This section reflects on the broader implications of the proposed suicide risk detection engine, its alignment with global sustainability goals, and the practical constraints of real-world deployment.

6.1 Alignment with UN Sustainable Development Goals

The United Nations created 17 Sustainable Development Goals (SDGs) [23] as a plan to build a better and more sustainable future for everyone by 2030. The development of this clinical AI engine directly contributes to these goals and we prioritize SDGs 3, 10, and 17 because they represent the intersection of clinical safety, equitable access, and regulatory governance. Focusing on these three ensures that the AI's impact is measurable, audit-ready, and directly aligned with the most urgent priorities of the Australian healthcare landscape.

6.1.1 Good Health and Well-being

Traditional screening tools often fail because they are too static to capture the nuances of a developing crisis, leading to what is known as "actuarial failure" where high-risk individuals are misclassified. By implementing a strict 100% Recall mandate as a technical requirement, our architecture transforms the AI into a reliable safety net. This ensures that the system directly reduces preventable mortality by identifying every high-acuity case and facilitating immediate clinical intervention.

6.1.2 Reduced Inequalities

The 8B parameter model is optimized to run on-premise, it can be deployed in regional and remote clinics that often lack specialized psychiatric staff or reliable high-speed internet. This is particularly vital for rural communities. By bringing high-tier detection capabilities to these underserved areas, the system bridges the geographical gap in mental health equity without the stigma often associated with traditional hospital admissions.

6.1.3 Partnerships for the Goals

The proposed model requires and enables cross-sector partnerships: AI startups, public health services, telehealth providers, regulatory bodies, and university research institutions. This creates a collaborative ecosystem for responsible clinical AI.

6.2 Carbon Footprint Analysis

To estimate the environmental impact of this work, the Machine Learning CO₂ Impact Calculator [24] was used to simulate the emissions generated during the development phase. The parameters for this simulation were carefully selected to reflect a professional, localized, and security-conscious clinical deployment. Hardware (A100 SXM4 80 GB) chosen as the industry standard for LLM fine-tuning, providing the necessary memory to handle the Llama 3.1 architecture efficiently. Region ap-southeast-2 - Sydney, selected to prioritize Australian Data Sovereignty (APP 8). Duration of 12 Hours that represents a realistic estimation for a LoRA-based fine-tuning cycle on a specialized clinical dataset.

Based on these parameters, the simulation yielded the following results: Experiments were conducted using Amazon Web Services (AWS) in region ap-southeast-2, which has a carbon efficiency of 0.8 kgCO₂eq/kWh. A cumulative of 12 hours of computation was performed on hardware of type A100 SXM4 80 GB (TDP of 400W). Total emissions are estimated to be 3.84 kgCO₂eq, of which 0 percent were directly offset by the cloud provider.

Moreover, the use of an 8B parameter model (Llama 3.1) with 4-bit quantization ensures that the system can run on local, low-energy medical servers. This avoids the massive energy overhead of routing every dialogue to a massive 70B+ model in a centralized data center.

6.3 Data Security and Privacy

Given the extreme sensitivity of psychiatric data, our design adheres strictly to the Privacy Act 1988 and Australian Privacy Principles (APP).

6.3.1 On-device Processing

All analysis is performed on local clinical hardware. Unlike cloud-based LLMs, the raw transcripts never leave the hospital's secure network, ensuring no cross-border data transfer (APP 8).

6.3.2 Data Management

Data is processed in-memory. If storage is required for long-term clinical records, all Personal Health Information (PHI) is de-identified using automated NLP filters before being committed to encrypted, AES-256 repositories.

6.4 Limitations

Despite the robust architecture, several limitations remain:

6.4.1 Linguistic Nuance

While the model is fine-tuned on the CRADLE Bench, it may still struggle with very specific local slang or non-standard English dialects not represented in the training corpus.

6.4.2 Context Windows

In extremely long, multi-day conversations, maintaining the full clinical context remains a technical challenge for an 8B model.

6.4.3 Hallucination Risks

Although the Hybrid RAG-FT architecture significantly reduces hallucinations, the risk of false confidence in low-risk scenarios still exists, requiring a clinician-in-the-loop at all times.

6.4.4 Demographic Bias

Despite our fine-tuning efforts, the foundational pre-training of Llama 3.1 remains English-dominant and reflects Western digital subcultures. This creates a risk where non-English speakers or communities with different cultural expressions of distress may be underrepresented, potentially reducing the model's recall for these specific populations.

6.5 Future Work

While the proposed system establishes a viable foundation for LLM-based suicide risk detection, several directions remain open for future investigation. These extensions target the key limitations identified in this work.

6.5.1 Australian-Specific Dataset Collection

A critical limitation of current benchmarks is their reliance on English-language Reddit data that does not reflect Australian linguistic and cultural diversity. Future work will focus on building a purpose-collected Australian corpus through partnerships with telehealth providers.

6.5.2 Multimodal Integration

Future iterations will explore integrating acoustic features (tone of voice, pauses) to complement the text-based linguistic markers for a more holistic risk assessment.

6.5.3 Edge Deployment

We aim to further optimize the model for Llama 3.2 1B/3B variants to allow for deployment on secure, mobile tablets for field workers in remote communities.

6.5.4 Longitudinal Tracking

Developing the capability for the AI to track a patient's risk trajectory over months, rather than just isolated sessions, to identify slow-building crises before they become imminent.

7 CONCLUSION

This research addressed the technical and ethical challenges of implementing a real-time suicide risk detection engine within the Australian healthcare context. The primary goal was to develop a system that prioritizes patient safety through high-sensitivity detection while maintaining clinical explainability and data sovereignty.

The benchmarking phase revealed a critical trade-off between the rapid inference of encoder models and the deep reasoning capabilities of decoder-based LLMs. While IBM Granite 3.2 showed the best ordinal alignment (0.545 QWK), Llama 3.1-8B emerged as the most versatile backbone for a

specialized pipeline. A key finding was the “domain shift” between social media data and clinical dialogue; models trained on Reddit style monologues required significant adaptation to handle the nuances of multi-turn conversational distress.

To bridge this gap, a hybrid architecture was proposed. By combining LoRA fine-tuning with Retrieval Augmented Generation (RAG), the system ensures that risk classifications are not only accurate but also grounded in verified Australian clinical standards (SAK). The proof-of-concept validation proved that the inclusion of Chain-of-Thought (CoT) reasoning and explicit safety constraints can raise the Recall for high-risk cases to 100%, a non-negotiable requirement for life-saving interventions.

Despite these advances, limitations such as linguistic bias and the management of long-term conversational context remain. Future work will focus on integrating multi-modal signals, such as acoustic markers, and optimizing the engine for edge deployment in regional and remote Australian communities.

In conclusion, this work demonstrates that lightweight, open-source LLMs can be successfully adapted for high-stakes mental health screening. By prioritizing transparency and regulatory compliance, this detection engine provides a scalable solution to support clinicians in identifying early warning signs, ultimately contributing to the reduction of preventable suicide deaths in Australia.

APPENDIX 1 - VIDEO PRESENTATION

A video demonstration of the work is available at: <https://echo360.net.au/media/6c58708d-0b8f-4fe3-981a-d7a969239302/public>

APPENDIX 2 – SUPPLEMENTARY MATERIALS

All supplementary materials, including cited papers, literature review notes, Python scripts, and additional figures, are available in the following Google Drive folder:

https://curtin-my.sharepoint.com/:f:/g/personal/21409109_student_curtin_edu_au/IgBGEzWfPZphR4kkSoBZnzTLAZb2C7kXM9tHd-Xf09m2BRQ?e=FV2bOW subdirectories:

- 01_Papers/ — Electronic copies of all cited papers.
- 02_Code/ — Jupyter notebooks and Python scripts used for benchmarking and evaluation (poc)
- 03_Extra/ — Extra materials and figures not included in the main report

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If you need to acknowledge anyone please include here.

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